



Department of Computer Science and Engineering (Data Science)

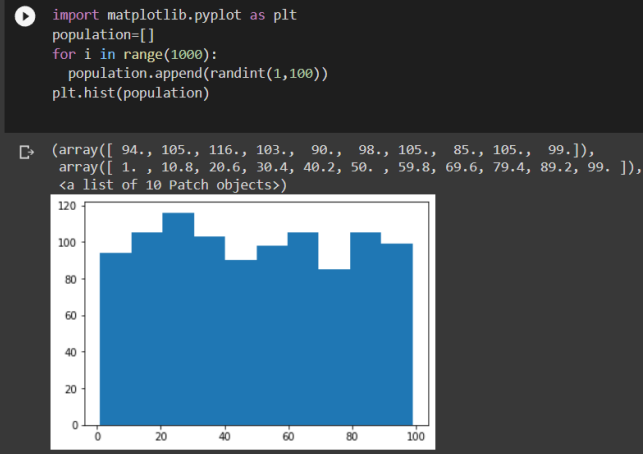
B.Tech. Sem: III Subject: Statistics for Data Science

Experiment 3

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SAP ID:60009210191

Date:	Experiment Title: Verify Central Limit Theorem
Aim	To verify Central Limit Theorem using Python
Software	Google Colab
Implementation	1. Generate random integer number in a range 0 to 100 using Python. Code: <pre>[1] from numpy.random import randint print(randint(0,100))</pre> 8 2. Generate floating point number in a range 0 to 100 using Python. Code: <pre>import random print(random.uniform(0,100))</pre> 16.17750780938629 3. Create a list of size 1000 random integer numbers in a range 0 to 100 and store in variable named as population using Python. Code: <pre>import random population= [] for i in range(0,1000): x = random.randint(0,100) population.append(x) print(population)</pre> [52, 21, 9, 34, 70, 50, 29, 57, 53, 0, 34, 39, 4, 80, 5, 95, 36, 11, 50, 27, 53, 18, 89, 31, 20, 1, 74, 38, 78, 68, 90, 75, 97, 5, 76, 9, 94, 91, 95, 52, 54, 69, 92, 41, 64, 32, 5, 96] 4. Construct a histogram of population created in Question3 using Python. Code:



5. Find mean and variance of population created in Question3 using Python.

Code:

```
import numpy as np
population=[]
for i in range(1000):
    population.append(randint(1,100))
print(np.mean(population))
print(np.var(population))
```

```
49.788
781.2790560000001
```

6. Draw a random sample of size n (user defined) from population created in Question3 using Python.

Code:

```
import numpy as np
a=int(input("enter a number "))
population=[]
for i in range(1000):
    population.append(randint(1,100))
print(random.sample(population,a))
```

```
enter a number 5
[90, 85, 23, 46, 3]
```

7. Create distribution of sample mean of samples drawn from population created in Question3 using Python. Here sample size and number of sample should be user defined.

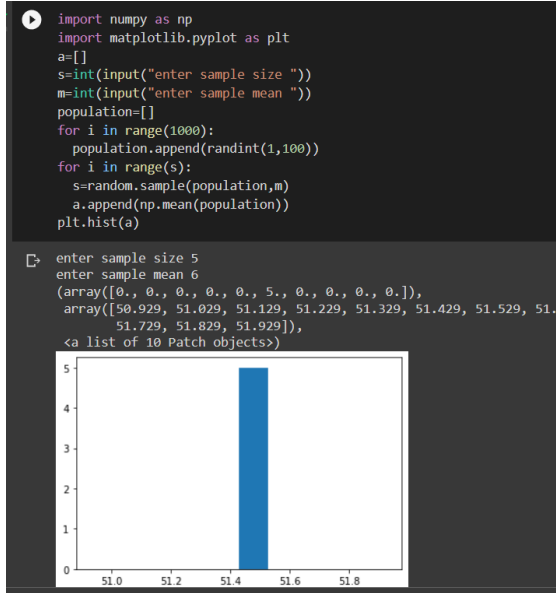
Code:

```
import numpy as np
s=int(input("enter a number for sample size "))
m=int(input("enter a number for sample mean "))
population=[]
for i in range(1000):
    population.append(randint(1,100))
for i in range(s):
    a=random.sample(population,m)
    print(a)
```

```
enter a number for sample size 5
enter a number for sample mean 6
[62, 42, 31, 37, 9, 90]
```

8. Construct a histogram of distribution of sample mean created in Question7 using Python.

Code:



9. Create a list of size 1000 floating point numbers in a range 0 to 100 and store in variable named as population using Python.

Code:

```
import random

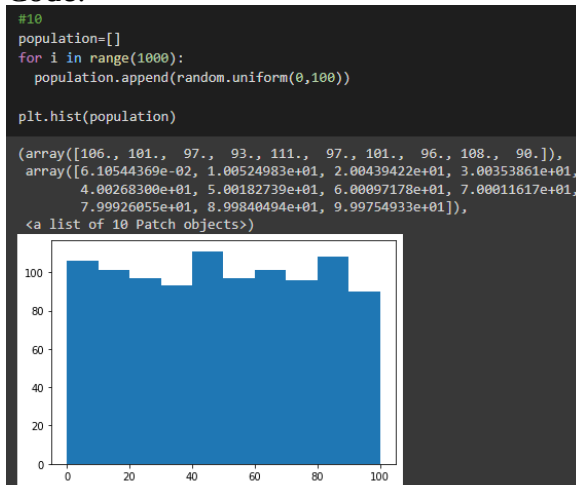
population= []
for i in range(1000):
    x = random.uniform(0,100)
    population.append(x)

print(population)
```

[86.90103573476335, 53.326433373215245, 91.13516479132404, 74.34948115120797, 62.84442948248182, 89.58476966420736, 87.91984687924406, 31.789522240320565, 21.732532695767294, 54.75

10. Construct a histogram of population created in Question9 using Python.

Code:



11. Find mean and variance of population created in Question9 using Python.

Code:

```
[22] population=[]  
      for i in range(1000):  
          population.append(random.uniform(0,100))  
      print(np.mean(population))  
      print(np.var(population))
```

```
51.330761818075175  
844.9894891486524
```

12. Draw a random sample of size n (user defined) from population created in Question9 using Python.

Code:

```
a=int(input("enter number"))  
population=[]  
for i in range(1000):  
    population.append(random.uniform(0,100))  
for i in range(100):  
    b=random.sample(population,a)  
    print(b)
```

```
enter number4  
[7.806991187410139, 78.44931522397451, 51.79484117764079, 27.642941489326013]
```

13. Create distribution of sample mean of samples drawn from population created in Question9 using Python. Here sample size and number of sample should be user defined.

Code:

```
[24] import numpy as np  
      s=int(input("enter a number for sample size"))  
      m=int(input("enter a number for sample mean"))  
      population=[]  
      for i in range(1000):  
          population.append(random.uniform(0,100))  
      for i in range(s):  
          a=random.sample(population,m)  
          print(a)
```

```
enter a number for sample size 4  
enter a number for sample mean 8  
[24.432302771283897, 12.258450032162616, 88.80104259194357, 61.049981706609046, 19.369167496569595, 68.97166092357222, 7.651754794977716, 49.97387664637372]
```

14. Construct a histogram of distribution of sample mean created in Question13 using Python.

Code:



15. Generate a normal population with user defined mean and standard deviation of size 1000 using Python.

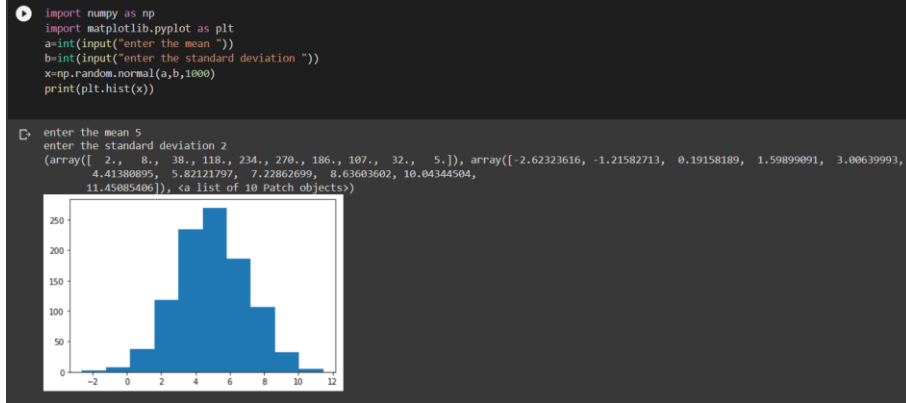
Code:

```
import numpy as np
a=int(input("enter the mean "))
b=int(input("enter the standard deviation "))
x=np.random.normal(a,b,1000)
print(x)
```

5.16325369e+00 6.67912263e+00 -2.11232340e+00 -7.72959574e-01
 2.67719947e+00 5.45814047e+00 3.49261455e-01 5.46694592e+00
 3.27080554e+00 4.67304339e+00 7.76738789e+00 2.75863355e+00
 3.37544099e+00 -2.72751954e+00 7.74044486e+00 7.63694742e+00
 3.13868292e+00 4.99576003e+00 1.11886937e+00 9.46174933e+00
 -2.89799909e+00 5.36843102e-01 6.17484042e-01 2.34874915e+00
 8.53894513e+00 4.30659572e+00 5.62786666e+00 3.35140436e+00
 6.16159172e+00 2.11042601e+00 4.91090891e+00 7.90115970e-01
 3.73267690e+00 1.87499407e+00 2.90354833e+00 1.06955643e+00
 4.67915892e+00 2.84448441e+00 2.81008513e+00 6.29813706e-01
 3.03158181e+00 5.42419293e+00 1.06405291e+00 3.72764118e+00
 5.68562970e+00 -2.10693591e+00 1.23364913e+00 6.41370255e+00
 4.94300262e-01 8.70401201e-01 -3.53268611e+00 4.87892030e+00
 6.29293627e+00 4.67155823e+00 2.91740712e+00 4.19270974e+00
 2.22463174e+00 7.97913723e+00 5.51995519e+00 5.39722454e+00
 4.41567965e-01 1.96054935e+00 1.35484936e+00 4.17359835e+00
 -8.14013897e-01 5.25402273e+00 4.32286503e+00 4.06585514e+00
 5.15945790e+00 7.19038035e+00 5.40980011e+00 3.07650371e+00
 6.07561873e+00 1.72542180e+00 2.51726223e+00 -4.97009102e+00
 -3.68655937e+00 2.66072997e+00 2.65317328e+00 2.17725710e+00
 4.96799770e+00 4.61380713e-02 4.49891797e+00 2.87769235e+00
 1.46102375e+00 5.91582989e+00 8.45131880e+00 2.79999264e-01
 2.16504627e+00 8.86960997e-01 3.05194396e+00 5.34692236e+00
 -2.22494261e-01 6.61792169e+00 -4.51144737e-01 4.67537108e+00
 8.62762962e+00 7.95218581e-01 -3.19884224e+00 -1.47351598e+00
 4.51258732e+00 3.11639623e+00 4.84764997e+00 -3.81709999e+00
 2.84137873e+00 1.08052572e+00 -1.96642715e+00 7.35211861e+00

16. Construct a histogram of population created in Question15 using Python.

Code:



17. Find mean and variance of population created in Question15 using Python.

Code:

```

[29] import numpy as np
import matplotlib.pyplot as plt
a=int(input("enter the mean "))
b=int(input("enter the standard deviation "))
x=np.random.normal(a,b,1000)
print(np.mean(x))
print(np.var(x))

```

enter the mean 4
 enter the standard deviation 5
 4.09777867173663
 24.990782881600207

18. Draw a random sample of size n (user defined) from population created in Question15 using Python.

Code:

```

[30] import numpy as np
import matplotlib.pyplot as plt
a=int(input("enter the mean "))
b=int(input("enter the standard deviation "))
c=int(input("enter a number "))
x=np.random.normal(a,b,1000)
for i in range(100):
    s=random.sample(tuple(x),c)
    print(s)

```

enter the mean 6
 enter the standard deviation 7
 enter a number 9
 [4.786046828584734, 1.56766067622139, -6.882839189575098, 3.1764171475224385, 5.6387216840588065, -0.9600264124093618, 1.17941456350544, 14.45871078832701, 3.2422900350917923]

19. Create distribution of sample mean of samples drawn from population created in Question15 using Python. Here sample size and number of sample should be user defined.

Code:

```

[31] import numpy as np
import matplotlib.pyplot as plt
import random
a=int(input("enter the mean "))
b=int(input("enter the standard deviation "))
c=int(input("enter sample size "))
d=int(input("enter a number of sample "))
x=np.random.normal(a,b,1000)
for i in range(d):
    s=random.sample(tuple(x),c)
    print(s)

```

enter the mean 6
 enter the standard deviation 8
 enter sample size 9
 enter a number of sample 4
 [10.435717420082785, 0.172018402928046, 5.573173589470505, -4.946293051921833, 4.8376901727153365, 6.204354714512037, 12.184679095810736, 2.9034009252964705, 4.177895508988542]

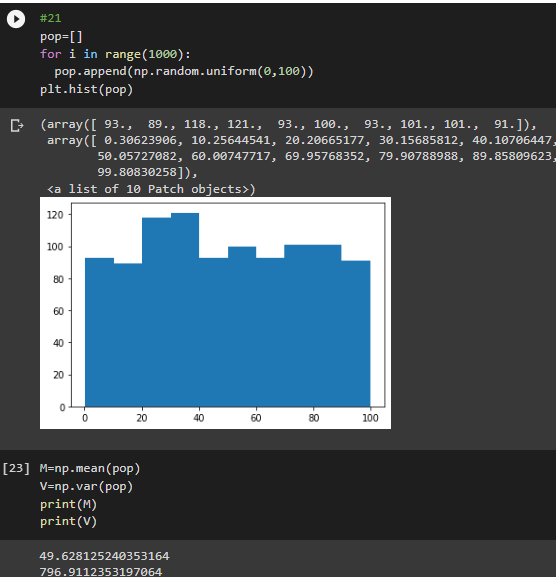
20. Construct a histogram of distribution of sample mean created in Question19 using Python.

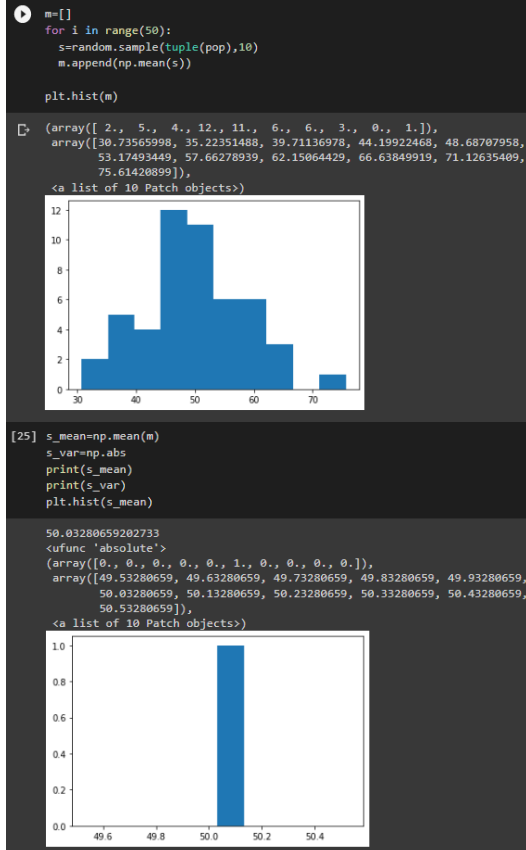
Code:



21. Verify Central Limit Theorem for Sampling distribution for Sample mean for randomly generated non-normal population using Python.

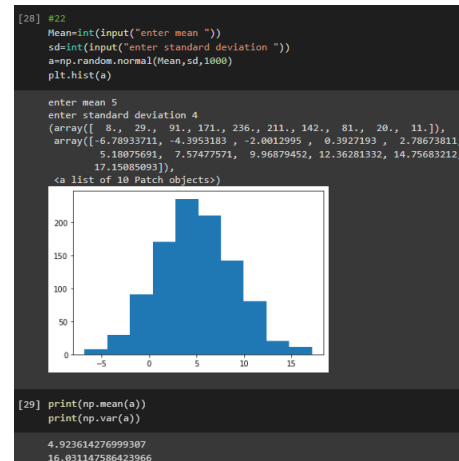
Code:





22. Verify Central Limit Theorem for Sampling distribution for Sample mean for randomly generated normal population using Python.

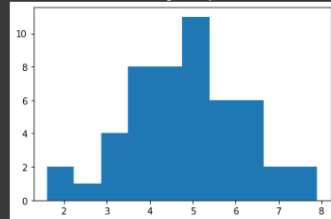
Code:




```
m=[]
for i in range(50):
    s=random.sample(tuple(a),10)
    m.append(np.mean(s))

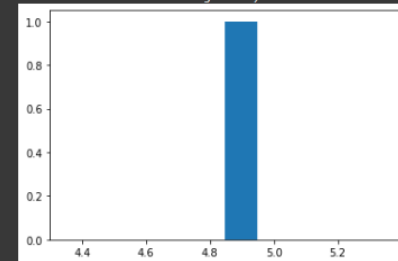
plt.hist(m)
```

```
[1: (array([ 2.,  1.,  4.,  8.,  8., 11.,  6.,  6.,  2.,  2.]),
      array([1.60967917, 2.23939729, 2.86911541, 3.49883354, 4.12855166,
            4.75826978, 5.3879879 , 6.01770602, 6.64742414, 7.27714227,
            7.90686039])),
      <a list of 10 Patch objects>)
```



```
[31] s_mean=np.mean(m)
      s_var=np.abs
      print(s_mean)
      print(s_var)
      plt.hist(s_mean)
```

```
4.8482190346640985
<ufunc 'absolute'>
(array([0., 0., 0., 0., 0., 1., 0., 0., 0., 0.]),
 array([4.34821903, 4.44821903, 4.54821903, 4.64821903, 4.74821903,
        4.84821903, 4.94821903, 5.04821903, 5.14821903, 5.24821903,
        5.34821903])),
      <a list of 10 Patch objects>)
```



23. We have population data for individual smoking habits. We know that the true population proportion for smoking is 0.395 or 39.5%. For the given data verify Central Limit Theorem for Sampling distribution for Sample proportions using Python. Take population size equal to 1000.

Code:

```
list_ones=list(np.ones(395,np.int))
list_zeros=list(np.zeros(505,np.int))
population=list_ones + list_zeros

fig, ax=plt.subplots(figsize=(10,7))
ax.hist(population, bins=3)

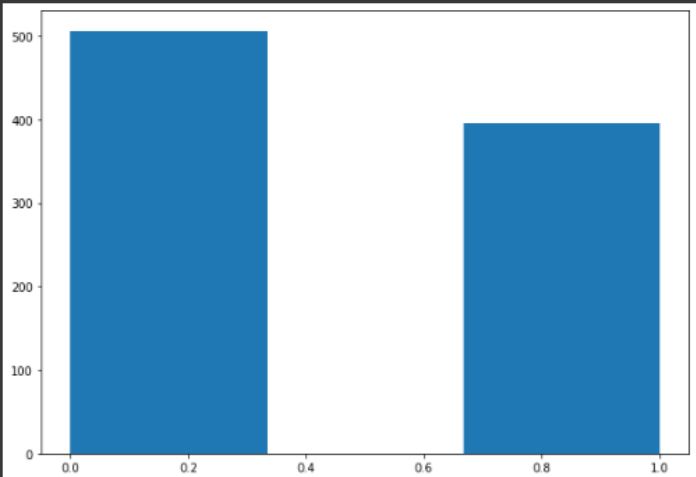
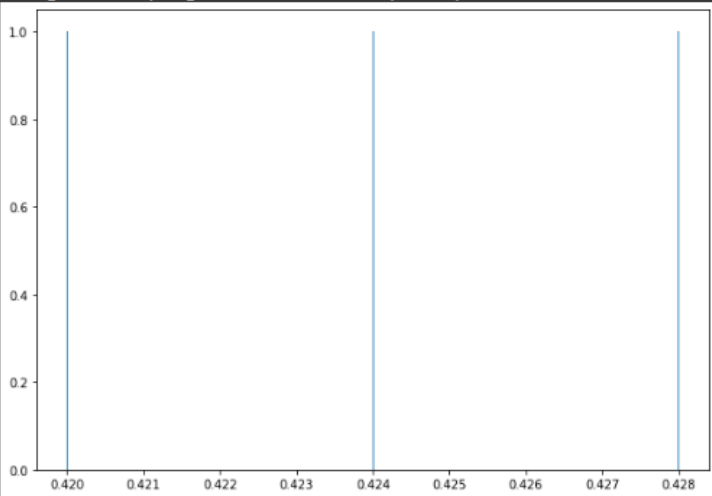
print("Histogram of Population Distribution is shown below:")
plt.show()

n= int(input("Enter the size of Sample you want to draw from your population:"))
m= int(input("Enter the number of sample you want to draw from your population:"))
sample_proportion=[]
for x in range(m):
    sample_data=random.sample(tuple(population),n)
    proportion=sum(sample_data)/n
    sample_proportion.append(proportion)
print("Histogram of sampling Distribution for Sample Proportion is shown below:")

fig, ax=plt.subplots(figsize=(10,7))
ax.hist(sample_proportion, bins=500)

plt.show()
```



	Histogram of Population Distribution is shown below: <pre>/usr/local/lib/python3.7/dist-packages/ipykernel_launcher.py:1: DeprecationWarning: Deprecated in NumPy 1.20; for more details and guidance: https://numpy.org/devdocs """Entry point for launching an IPython kernel. /usr/local/lib/python3.7/dist-packages/ipykernel_launcher.py:2: DeprecationWarning: Deprecated in NumPy 1.20; for more details and guidance: https://numpy.org/devdocs</pre>  Enter the size of Sample you want to draw from your population:250 Enter the number of sample you want to draw from your population:3 Histogram of sampling Distribution for Sample Proportion is shown below: 	
Conclusion	Central Limit Theorem is verified using various functions & libraries in python.	

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